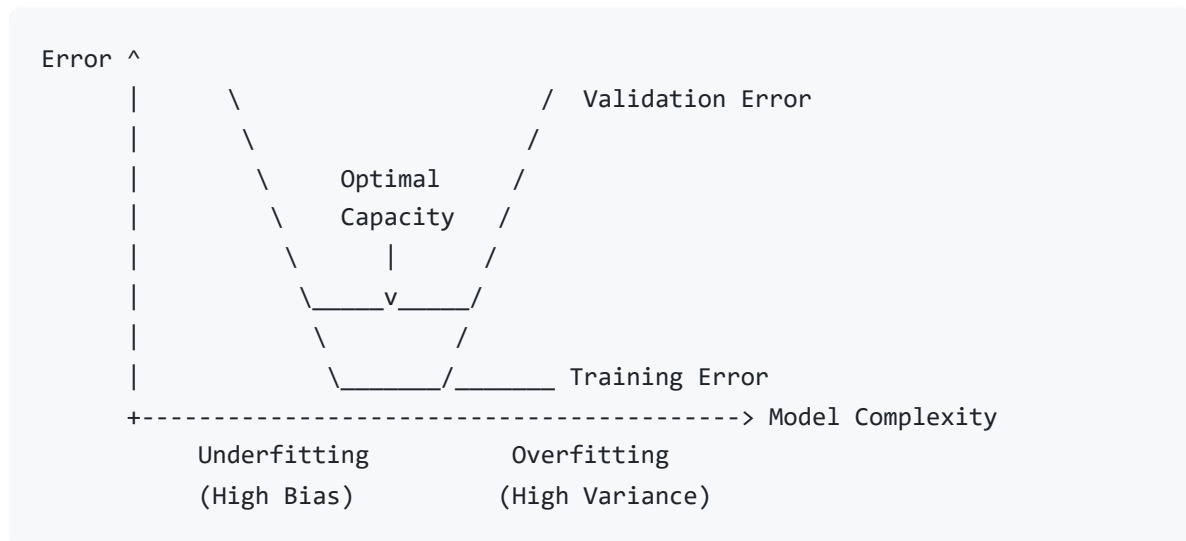


Module 24.06: How to Fix Underfitting & Overfitting

1. The Engineering Balance: Bias-Variance Optimization

Machine learning practitioners must calibrate model capacity to find the sweet spot where total generalization error is minimized:



2. Concrete Strategies to Mitigate Underfitting (High Bias)

When a model suffers from underfitting ($\mathcal{L}_{\text{train}}$ is high):

Action	Technical Implementation	Why it Works
1. Increase Model Capacity	Use non-linear models (Polynomial, Splines, Trees, Neural Nets)	Expands hypothesis space $\mathcal{H}$
2. Feature Engineering	Add polynomial powers (x^2, x^3) and interaction terms ($x_1 x_2$)	Projects input into higher-dimensional linear separable space
3. Decrease Regularization	Reduce penalty λ (decrease α in Lasso/Ridge)	Allows parameter weights $\mathbf{w}$ more freedom to fit data
4. Train for Longer	Increase iterations, tune learning rate α	Ensures optimizer reaches convergence minimum

3. Concrete Strategies to Mitigate Overfitting (High Variance)

When a model suffers from overfitting ($\mathcal{L}_{\text{val}} \gg \mathcal{L}_{\text{train}}$):

Action	Technical Implementation	Why it Works
1. Apply Regularization	Add L1 (Lasso) or L2 (Ridge) penalty to cost function	Constrains weight norms, preventing explosive coefficients
2. Acquire More Data	Data collection, data augmentation	Constrains variance; sample distribution converges to population
3. Dimensionality Reduction	Feature selection, PCA, drop noise columns	Eliminates spurious chance correlations in high dimensions
4. Cross-Validation	K-Fold cross validation, early stopping	Halts training before validation loss begins to diverge
5. Simplify Model Architecture	Reduce polynomial degree, prune decision trees	Restricts hypothesis complexity to match true signal complexity